Advanced System Programming

Note Application using KV Store

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RATATUI

Cook up delicious terminal user interfaces.

Ratatui is a Rust library for building fast, lightweight, and rich terminal user interfaces 👨‍🍳 🐀

Get Started | Browse Examples | API Docs

```rust
use ratatui::style::Stylize;
use ratatui::widgets::{Block, Paragraph};

fn main() -> Result<(), Box<dyn std::error::Error>> {
    ratatui::run(|terminal| {
        terminal.draw(|frame| {
            let block = Block::bordered().title("Welcome");
            let greeting = Paragraph::new("Hello, Ratatui! 🐭")
                .centered()
                .yellow()
                .block(block);
            frame.render_widget(greeting, frame.area());
        })?;
        std::thread::sleep(std::time::Duration::from_secs(5));
        Ok(())
    })
}
```

0.30.0

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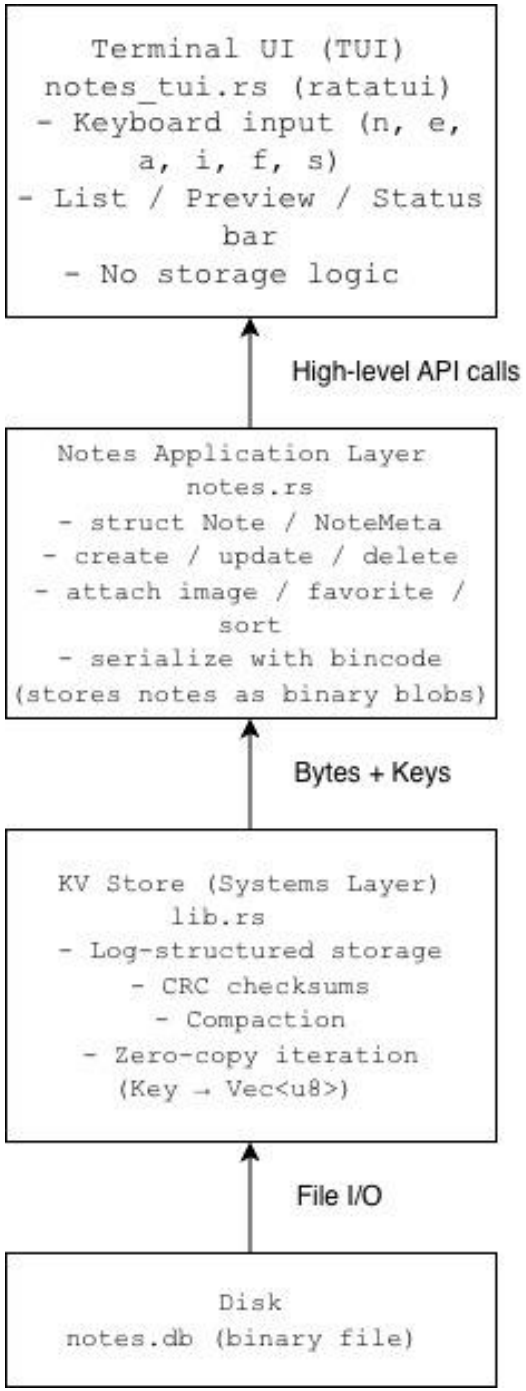
Rust library used for building rich, fast, and lightweight terminal user interfaces (TUIs) and interactive console applications. It enables developers to create complex, text-based dashboards, interfaces, and command-line tools with widgets like layouts, tables, and graphs, often using Crossterm for input handling.


Terminal UI (TUI)
notes_tui.rs (ratatui)
- Keyboard input (n, e, a, i, f, s)
- List / Preview / Status bar
- No storage logic
High-level API calls
Notes Application Layer
notes.rs
- struct Note / NoteMeta
- create / update / delete
- attach image / favorite / sort
- serialize with bincode
(stores notes as binary blobs)
Bytes + Keys
KV Store (Systems Layer)
lib.rs
- Log-structured storage
- CRC checksums
- Compaction
- Zero-copy iteration
(Key → Vec<u8>)
File I/O
Disk
notes.db (binary file)